Decision Trees, Bagging, and Multiclass Boosting on Olivetti Faces

Machine Learning Assignment 2 Report

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Repo Link:
https://github.com/Chandu-Suggu/ML_Assignment2

Abstract. This report combines the three supplied assignment notebooks: decision-tree selection and bootstrap aggregation (Q1), a from-scratch multiclass SAMME boosting implementation (Q2), and repeated-resampling estimates of bias and variance (Q3). The notebooks use the Olivetti Faces dataset, comprising 400 grayscale images represented as 4,096 flattened pixel features. Saved outputs report a 240/80/80 train/validation/test split. Q1 records a 100-tree bagged ensemble with 88.75% clean test accuracy and 87.50% masked test accuracy, exceeding its selected single tree. Q2 reports lower performance for the selected SAMME model, and Q3 reports low measured variance but high squared bias for boosting. Bagging is the strongest method in the recorded Q1 and Q3 outputs. However, some model settings and metrics differ between notebooks, so the report preserves results by question instead of claiming all figures came from one identical run.

1. Introduction

A single decision tree can fit complex pixel patterns, but its decision boundary may change substantially when the training sample changes. Bagging fits trees to bootstrap samples and aggregates their predictions, primarily reducing variance. Boosting instead trains a sequence of weak learners and increases the relative influence of examples misclassified by earlier learners. These mechanisms can yield different accuracy, confidence, and robustness profiles.

The assignment evaluates tree complexity and bagging (Q1), implements multiclass SAMME (Q2), and estimates measured bias and variance over repeated resampling (Q3). The report treats the supplied executed notebook outputs as the source of numerical results.

2. Dataset and Experimental Protocol

The notebooks load fetch_olivetti_faces: 400 grayscale images of size 64 × 64, each flattened into 4,096 pixel features. No PCA or dimensionality reduction is used in the described pipeline. Saved executions report 240 training, 80 validation, and 80 test examples. The intended split is identity-wise: six images per identity for training, two for validation, and two for testing, with the numeric roll-number seed. The clean test set should remain fixed across experiments.

For robustness evaluation, the test image region at rows 16–31 and columns 24–39 is set to zero. Models are trained on clean images only. Validation accuracy is used for model selection, and test data are reserved for final evaluation.

> **Reproducibility caveat.** Recorded settings differ: Q1 selects an unlimited-depth tree with minimum leaf size 2; Q2 selects SAMME depth 3 with 100 rounds; Q3 reports depth 4 with 50 rounds. Some test metrics also differ between earlier and later output cells. These values are reported as recorded and should be regenerated under one frozen configuration before final submission.

3. Q1: Decision Tree and Bagging

3.1. Single-tree selection

The grid search maximum depths {2, 4, 8, 16, None} and minimum leaf sizes {1, 2, 4}. Q1 selects max_depth=None and min_samples_leaf=2 using validation accuracy. The saved metrics are 99.17% training, 67.50% validation, and 62.50% clean test accuracy. The selected tree has 43 leaves, approximately 5.58 training examples per leaf on average, and no single-example leaves. The large train–validation gap suggests overfitting. Shallow trees are strongly underfit for the 40-class identity task, while deeper trees fit training data more closely.

3.2. Bootstrap aggregation

Bagging draws training examples with replacement, fits one decision tree per bootstrap sample, and aggregates class votes. A bootstrap sample of size 240 contains about 151.7 distinct examples in expectation, matching the recorded distinct-sample counts near 151–158. Q1 chooses B=100 trees by validation performance.

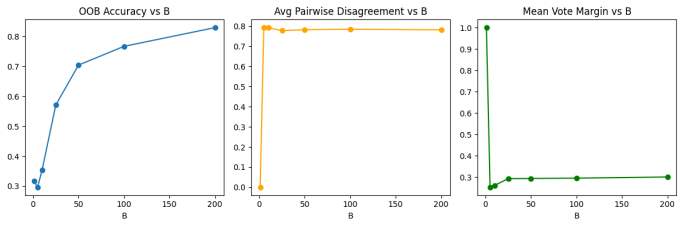
B	Train	Val	Test	Avg Distinct Samples
1	0.7250	0.3250	0.4500	152.00
5	0.9542	0.5375	0.5875	152.00
10	1.0000	0.6750	0.7000	153.10
25	1.0000	0.8125	0.8000	153.28
50	1.0000	0.8250	0.8500	152.84
100	1.0000	0.8500	0.8750	152.44
200	1.0000	0.8625	0.8875	152.39

Table 1. Q1 saved bagging sweep. The 100-tree ensemble improves clean test accuracy from 62.50% for the selected tree to 88.75%, a 26.25 percentage-point gain. Increasing the ensemble to 200 trees does not improve validation or test accuracy, suggesting diminishing returns.

3.3. OOB diagnostics and robustness

The notebook examines out-of-bag (OOB) accuracy, average pairwise tree disagreement, and vote margin as ensemble size changes. OOB predictions use only trees for which a training example was omitted from the bootstrap sample. Disagreement measures diversity, while vote margin is the winning vote fraction minus the runner-up fraction. Diversity is useful only when the constituent trees remain sufficiently accurate.

Q1 reports single-tree accuracy of 62.50% on clean images and 60.00% on masked images. Bagging scores 88.75% clean and 87.50% masked. Its mean confidence decreases from 0.3820 to 0.3030 under masking even though accuracy drops by only 1.25 percentage points. The ensemble is more robust in this run, although confidence indicates that the mask still removes useful evidence.



4. Q2: Multiclass SAMME Boosting

4.1. Update rule

For K classes, SAMME uses $\alpha_t = \log((1-\varepsilon_t)/\varepsilon_t) + \log(K-1)$, where ε_t is the weighted classification error of weak learner t. Class score $S_t(x) = \sum_t \alpha_t \cdot 1[h_t(x)=k]$; prediction selects the class with the largest score. Samples misclassified by a weak learner receive increased relative weight. Implementations should explicitly handle learners whose weighted error is no better than random multiclass prediction.

Metric	Value
Initial weights	[0.1667, 0.1667, 0.1667, 0.1667, 0.1667, 0.1667]
Predictions	[0, 1, 1, 2, 2, 0]
True Labels	[0, 0, 1, 1, 2, 2]
Weighted Error	0.5000
Learner Weight (a_t)	0.6931
Updated Weights	[0.1111, 0.2222, 0.1111, 0.2222, 0.1111, 0.2222]

4.2. Model selection and comparison

Q2 selects SAMME depth 3 with 100 rounds at validation accuracy 40.00%. The recorded round-count comparison is shown in Table 2.

Rounds	Model	Train Acc	Val Acc	Test Acc	Time
25	Bagging	0.512	0.388	0.362	22.914s
25	Boosting	0.400	0.150	0.163	11.944s
50	Bagging	0.675	0.487	0.400	45.236s
50	Boosting	0.713	0.300	0.263	23.271s
100	Bagging	0.704	0.487	0.475	91.009s
100	Boosting	0.871	0.400	0.300	46.559s

Table 2. Q2 recorded comparison; rounded from saved outputs. At 100 rounds, boosting reaches 87.1% training accuracy but only 30.0% test accuracy, a large generalization gap. Bagging reaches 47.5% test accuracy in this comparison. The 50-round bagging run has the highest listed validation and test accuracy, although boosting's selected parameters come from its separate validation sweep.

4.3. Hard examples and masked data

The Q2 output identifies training image index 47 as the heaviest sample, with final weight 0.0374; its prediction history is mostly incorrect over the first 25 rounds. Boosting focuses on difficult examples, but this can overemphasize ambiguous or noisy samples. The saved robustness output reports bagging at 43.75% clean and 50.00% masked accuracy, with mean confidence 0.1711. Boosting scores 30.00% clean and 26.25% masked, with confidence around 0.1007 and 0.1004. A highlighted masked example also shows different class choices by the two ensembles; one example is illustrative rather than statistically conclusive.

Model	Condition	Accuracy	Mean Confidence
Bagging	Clean	0.4750	0.1847
Bagging	Masked	0.4250	0.1847
Boosting	Clean	0.3000	0.1007
Boosting	Masked	0.2625	0.1004

5. Q3: Measured Bias and Variance

5.1. Estimation protocol

Q3 forms 30 resampled training sets by sampling within each identity, then refits the selected tree, bagged model, and boosting model without retuning. Let $p_r(x_i)$ be a model's class-score/probability vector for test example i in repetition r, and $\bar{p}(x_i)$ the mean across repetitions. Measured squared bias is the mean squared Euclidean distance between $\bar{p}(x_i)$ and the one-hot target e_i. Variance is the average squared distance between $p_r(x_i)$ and $\bar{p}(x_i)$.

Model	Clean Test Acc (Mean)	Masked Test Acc	Measured Bias^2	Measured Variance
Single Tree (Selected)	0.487083	0.3125	0.377389	0.648444
Bagging (Selected)	0.776667	0.8250	0.167833	0.278833
Boosting (Selected)	0.272500	0.3625	0.883492	0.020592

5.2. Summary results

Model	Clean Test Acc (Mean)	Masked Test Acc	Measured Bias^2	Measured Variance
Single Tree (Selected)	0.487083	0.3125	0.377389	0.648444
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Table 3. Q3 recorded repeated-resampling summary. The single tree has high measured variance, consistent with sensitivity to sampled training faces. Bagging reduces both measured squared bias and variance and has the highest reported mean accuracy. Boosting has the lowest variance but

the largest squared bias and lowest accuracy: predictions are stable across resamples but remain far from the one-hot targets. Low variance alone therefore does not imply good classification performance.

5.3. Effect of tree depth

For depth 2, the notebook reports tree bias² 1.7976 and variance 0.0857, while bagging has bias² 0.8759 and variance 0.8541, producing a negative estimated variance reduction (−0.7684). For unlimited depth, tree variance is 0.6484 and bagged variance is about 0.259, a reduction of roughly 0.39. These are finite-resampling estimates. The results suggest bagging is most helpful when the base learner is unstable; shallow, biased trees may not benefit similarly.

The deepest tree is not automatically the depth at which bagging yields the greatest accuracy improvement. Depth interacts with bias, variance, leaf occupancy, and tree disagreement. Deep trees can be unstable and offer variance to average away, but correlated or weak trees limit the gain. Shallow trees can have high bias that aggregation cannot remove.

Depth	Tree Bias-square	Tree Var	Bag Bias-square	Bag Var	Var Removed
2	1.7976	0.0857	0.8759	0.8541	-0.7684
None	0.3774	0.6484	0.1402	0.2590	0.3895
None	0.3774	0.6484	0.1482	0.2585	0.3900

7. Limitations and Conclusion

The dataset contains only 400 images and six training examples per identity, and raw pixels provide no learned invariance to pose or illumination. The fixed rectangular mask cannot represent all real occlusions. Uncertainty intervals and statistical tests are not reported for every comparison. Most importantly, notebook outputs contain inconsistent settings and metrics. The recorded evidence nevertheless suggests bagging is the strongest method in Q1 and Q3: it improves clean accuracy, retains performance under the specified mask, and reduces measured bias and variance relative to the tree. The SAMME example demonstrates reweighting, but its recorded face-classification accuracy is weaker and its training–test gap is large. Final claims should be based on a reproducible rerun.

References

[1] L. Breiman, “Bagging predictors,” Machine Learning, vol. 24, pp. 123–140, 1996.

[2] J. Zhu, S. Rosset, H. Zou, and T. Hastie, “Multi-class AdaBoost,” Statistics and Its Interface, vol. 2, no. 3, pp. 349–360, 2009.

[3] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, Classification and Regression Trees. Wadsworth, 1984.

[4] Scikit-learn developers, “Olivetti faces dataset,” scikit-learn documentation, fetch_olivetti_faces.

[5] Assignment 2 materials: ASSIGNMENT_2_1.ipynb, ASSIGNMENT_2_2.ipynb, ASSIGNMENT_2_3.ipynb, and supplied specification images. Numerical values were transcribed from saved notebook outputs.